Left Truncation

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Introduction

Left truncation, also called delayed entry, occurs when subjects become observable only after surviving to a certain age or calendar time. Recruiting middle-aged adults into a cohort study means anyone who died as a child or young adult is structurally absent — they could not contribute. Treating delayed-entry data as if every subject had been followed from time zero overestimates survival, because the early period is observed only for the survivors. The counting-process formulation in `survival::Surv(tstart, tstop, event)` handles this naturally by restricting risk sets at each event time to subjects actually under observation then.

Prerequisites

A working understanding of survival analysis, the difference between censoring (subject leaves before event) and truncation (subject only enters under specific conditions), and the counting-process formulation of survival data.

Theory

For a left-truncated subject who enters at time τ_i and exits at time T_i (with event indicator δ_i), the contribution to the partial likelihood at event time t should appear only when $\tau_i < t \leq T_i$. The standard $\text{Surv}(\text{time}, \text{event})$ object does not encode the entry time τ_i and silently sets it to zero, biasing estimates upward. The two-argument form $\text{Surv}(\tau, T, \text{event})$ (counting-process style) carries both endpoints; `survfit()` and `coxph()` use them to construct correct risk sets.

In age-as-timescale analyses (the natural framing in chronic-disease epidemiology), every subject is left-truncated at their age at enrolment.

Assumptions

The truncation mechanism is independent of the event of interest given covariates: subjects do not become observable for reasons related to their underlying risk profile.

R Implementation

```r
library(survival)

# Age-as-timescale cohort: enrolled between ages 40 and 70
set.seed(2026)
n <- 200
age_enroll <- runif(n, 40, 70)
event_age  <- age_enroll + rexp(n, 0.05)
end_age    <- pmin(event_age, age_enroll + runif(n, 5, 20))
status     <- as.integer(event_age <= end_age)

# Left-truncated Cox: tstart = entry age, tstop = exit age
fit <- coxph(Surv(age_enroll, end_age, status) ~ 1)
```

```r
summary(fit)

# Kaplan-Meier with delayed entry
km <- survfit(Surv(age_enroll, end_age, status) ~ 1)
plot(km)
```

Output & Results

`survfit()` returns a Kaplan-Meier curve on the chosen time scale (age, calendar year, or time-since-event) that conditions on each subject's entry time. The Cox model with the counting-process `Surv` object produces unbiased hazard-ratio estimates even when entry times are heavily truncated.

Interpretation

A reporting sentence: "Using age as the time scale with left truncation at enrolment, the conditional survival from age 40 was 0.75 at age 60 (95 % CI 0.68 to 0.82); ignoring delayed entry inflates this estimate to 0.84 by treating mid-life enrolees as if they had been followed since age 0." Always state the time scale and the truncation mechanism in the methods section.

Practical Tips

- Always specify both `tstart` and `tstop` arguments to `Surv()` when entry times vary across subjects; the single-argument form silently assumes all subjects entered at zero.
- Age-as-timescale cohort studies are always left-truncated; this is the most common scenario in chronic-disease epidemiology and ageing research.
- Distinguish left truncation (subject only observable after a certain time) from left censoring (event occurred before observation began but the subject is observable now); the two are handled differently.
- Without left-truncation correction, you overestimate survival of cohorts enrolled at older ages because their early-life mortality is invisible to your data.
- `survfit()` with delayed entry produces correct conditional survival curves; check that the reported time scale starts at the minimum entry time, not zero.
- For competing risks with delayed entry, `cmprsk::cuminc` and `tidycmprsk::cuminc` accept counting-process input via `entry` arguments.

R Packages Used

`survival` for `Surv(tstart, tstop, event)` and `survfit()` / `coxph()` with counting-process input; `Epi` for left-truncation-aware tools targeted at epidemiology; `etm` for non-parametric multi-state estimation with delayed entry; `tidycmprsk` for left-truncation-aware competing-risks analysis.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models

- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation